Positional Information for Budget-Adaptive Markov Prediction

Local finite-horizon research draft, 2 November 2026

Abstract

Consider prediction from one binary order-two Markov trajectory after an unknown number of symbol edits, with the terminal pair protected. The loss is expected total variation at that actual pair. At a supplied all-state two-step minorization certificate $1/M$, the known-budget risk scale is $\max\{M^{-1/2}, k/M\}$. We ask which restricted observations preserve this scale simultaneously for every budget k. For every finite horizon $M \geq 64$ and integer $2 \leq r \leq M+2$, the optimal budget-blind competitive factor from exact triple counts and the first and last r symbols lies between $2^{-30}\ell$ and $2^{20}\ell$, where $\ell = 1+\log((M+2)/r)$. The upper uses only counts and the suffix; the lower permits both windows and compares the same map across budgets, rather than imposing a logarithmic penalty at every budget. Consequently a prescribed width sequence admits a finite uniform factor exactly when $\liminf_M r_M/M > 0$. A separate fixed-fraction argument retains stronger explicit lower constants. The positive construction accepts an empirical row estimate only when enough visits occurred before the observed suffix. The lower combines an occupation mixture with a count-preserving interior rearrangement and a pointwise legal edit kernel. Under a different access model, a whole-word scan can compress the needed gate into one exact supplemental bit. These are statistical observation results, not learning or architectural separation claims.

1 The observation question

A transition histogram records which local patterns occurred but not when they occurred. For a clean homogeneous Markov trajectory, this is consistent with the usual likelihood factorization into transition counts and the initial state. With position-sensitive corruption, however, the observation channel changes. A predictor may need to distinguish an empirical row supported by older visits from one populated almost entirely near the prediction time. The question here is how much positional information is needed when the number of edits is not supplied.

We study one concrete experiment: a binary order-two source, M observed departures, and an unmodified terminal pair. The source obeys an all-state two-step minorization with supplied coefficient $1/M$. An offline kernel may change at most k earlier symbols on every realization. The predictor sees exact triple counts and prescribed boundary windows, but not k. Its risk is expected unsquared total variation to the next-symbol distribution at the actual terminal pair. In this experiment, giving k to the predictor makes counts and the protected pair sufficient for the optimal *rate*; no claim of statistical sufficiency is needed.

Our main result characterizes the competitive factor simultaneously for every finite horizon and boundary width: its order is $1 + \log((M+2)/r)$ with universal constants. One map at each fixed horizon must serve every budget. The positive lower limiting fraction criterion and logarithmic fixed-fraction price follow at the order level; a separate fixed-fraction proof supplies a stronger explicit lower constant. A separate corollary shows why raw access and computed messages must

not be identified: a whole-word scan can produce a single exact gate bit that suffices beyond counts. The scan, exact counters, and protected query are part of that corollary's information model.

The proof mechanisms are complementary. For the upper bound, a gate retains evidence of visits before a suffix, removing the shortest-return terms from a harmonic convolution. For the lower bound, clean alternatives with small prediction risk are transformed into legally edited null words whose restricted observations match those alternatives. The clean guarantee is used at the same horizon and supplied certificate as the attacked guarantee. Complete arguments appear in the supplement; the main text isolates the experiment and the information comparisons.

2 Experiment and benchmark

All logarithms are natural. Let $M \geq 64$ be an integer and

$$X = (X_0, \ldots, X_{M+1}), \qquad Y_i = (X_i, X_{i+1}), \quad 0 \leq i \leq M.$$

Thus there are $M + 2$ raw symbols, M departures, and the terminal pair Y_M. A homogeneous order-two transition table $\theta_{ab}(c)$ induces the four-state pair kernel $P((a, b), (b, c)) = \theta_{ab}(c)$. There is a qualifying stationary law Π such that

$$\Pi P = \Pi, \qquad P^2(u, v) \geq \beta \Pi(v) \quad \text{for all } u, v \in \{00, 01, 10, 11\}. \tag{2.1}$$

The condition includes rows outside stationary support. Throughout the main results the supplied certificate is *exactly* $\beta = 1/M$; it need not be the source's largest possible coefficient. Neither θ nor Π is revealed. Stationary initialization means $Y_0 \sim \Pi$; arbitrary initialization means any hidden law μ, including laws supported outside the stationary support.

Strict edits and actual-terminal loss. For $0 \leq k \leq M$, a legal randomized kernel $K_{M,k}$ maps full binary words to full binary words. On every input x, every output z in its support satisfies

$$z_M = x_M, \qquad z_{M+1} = x_{M+1}, \qquad \sum_{i=0}^{M-1} \mathbf{1}\{z_i \neq x_i\} \leq k. \tag{2.2}$$

The kernel is one fixed map on words, not supplied the true source table, stationary or initial law, or predictor's random seed. It may inspect the whole input and use public parameters and k; lower-bound kernels may hardcode a predeclared auxiliary family. No causal or computational restriction is imposed. Its randomness is independent of the predictor's. This formulation specifies the common kernels used in the proofs; source independence alone is not asserted to distinguish all standard contamination models.

For $Z \sim K_{M,k}(X, \cdot)$ and an estimator A based on observation $\mathcal{T}(Z)$, define

$$R(A; \theta, \mu, K_{M,k}) = \mathbb{E}\ \mathrm{TV}\big(A(\mathcal{T}(Z)), \theta_{Y_M}\big). \tag{2.3}$$

The expectation includes the path and both independent randomizations. The target is the source's next-symbol law at the *actual protected pair*, not an independent query. For binary distributions, TV is absolute error in either coordinate. Risk is unconditional.

Observed information. For $a, b, c \in \{0, 1\}$ let

$$H_{abc}(z) = \sum_{i=0}^{M-1} \mathbf{1}\{(z_i, z_{i+1}, z_{i+2}) = (a, b, c)\}.$$

For prescribed $2 \le r \le M + 2$, the observation is

$$\mathcal{T}_r(z) = \big((H_{abc}(z))_{a,b,c},\ z_0, \dots, z_{r-1},\ z_{M+2-r}, \dots, z_{M+1}\big). \tag{2.4}$$

The windows may overlap. Counts and raw symbols are exact; neither edit locations nor the clean interior are supplied. In row $u = (a, b)$ use the designated successor $d(u) = 1 - a$, and define

$$J'_u = H_{ab0}(Z) + H_{ab1}(Z), \qquad C'_u = H_{ab,d(u)}(Z), \qquad q_u = \theta_{ab}(d(u)).$$

Primes indicate corrupted counts. Unprimed counts are clean analysis variables. At the selected row $u = Y_M$, subscripts are omitted. A ratio C/J is evaluated only when $J > 0$.

One family for all budgets. Use the explicitly normalized rate profile

$$g_M(k) = \max\{M^{-1/2}, k/M\}, \qquad 0 \le k \le M. \tag{2.5}$$

For a prescribed sequence $r_\bullet = (r_M)_{M \ge 64}$, let $C_{\mathrm{stat}}(r_\bullet)$ be the infimum of constants $C \ge 0$ for which one estimator family (A_M), based on $\mathcal{T}_{r_M}$, satisfies

$$R(A_M; \theta, \Pi, K_{M,k}) \le C g_M(k) \tag{2.6}$$

for every $M \ge 64$, every $0 \le k \le M$, every qualifying source and every legal kernel. Define C_{all} by additionally quantifying over every μ. The family may depend on the prescribed sequence, M, r_M, and supplied $1/M$, but not on k or hidden source quantities. The infimum need not be attained; an empty admissible set has value $+\infty$.

Finite-horizon competitive factors. Separately from the sequence factors C, fix (M, r) and a single k-blind map A based on $\mathcal{T}_r$. Define

$$\Gamma_{\mathrm{stat}}(A; M, r) = \sup_{\substack{0 \le k \le M,\ \theta, \Pi, K_{M,k} \\ \text{qualifying stationary experiments}}} \frac{R(A; \theta, \Pi, K_{M,k})}{g_M(k)}, \qquad \Gamma_{\mathrm{stat}}(M, r) = \inf_A \Gamma_{\mathrm{stat}}(A; M, r). \tag{2.7}$$

Define $\Gamma_{\mathrm{all}}(A; M, r)$ and $\Gamma_{\mathrm{all}}(M, r)$ by also allowing every initial law μ. The map may depend on M, r and the supplied $1/M$, but the very same map serves all k at that horizon. These are finite-horizon infima, not the sequence quantities C above; no infimum is assumed attained. Every map has factor at most $\sqrt{M}$ because TV is at most one.

The comparison is to a genuine same-information known-budget scale, rather than to an arbitrary denominator. Write $R^{\mathcal{T}}_{\mathrm{stat}}(M, k)$ and $R^{\mathcal{T}}_{\mathrm{all}}(M, k)$ for the respective minimax risks when k is also supplied.

Proposition 2.1 (Known-budget scale)**.** *For the full corrupted history, for $\mathcal{T}_r$, and for counts plus the protected pair,*

$$\frac{g_M(k)}{96} \le R^{\mathcal{T}}_{\mathrm{stat}}(M, k) \le R^{\mathcal{T}}_{\mathrm{all}}(M, k) \le 191488 g_M(k). \tag{2.8}$$

The upper is count-only and allows arbitrary starts; the lower holds even for full histories. Thus a factor C relative to g gives a factor at most $96C$ relative to the corresponding minimax oracle. Conversely, a factor D relative to that oracle gives a factor at most $191488D$ relative to g. Proofs are in Lemma B.5 and Proposition A.10.

For comparison with a more general certificate-dependent profile, define $F_2(M, 0, \beta) = 0$ and, for $k \geq 1$,

$$F_2(M, k, \beta) = \frac{k}{M}\left[1 + \log\left(1 + \frac{1}{k \max\{\sqrt{\beta}, 1/M\}}\right)\right], \qquad H_2 = \min\{1, \max\{M^{-1/2}, F_2\}\}.$$

At the certificate used here, $\log(1+x) \leq x$ gives $g_M(k) \leq H_2(M, k, 1/M) \leq 2g_M(k)$. Hence the two normalizations change optimal factors by at most two; all constants below are stated directly for g.

3 Finite-horizon characterization and its consequences

Theorem 3.1 (Simultaneous finite-horizon window price)**.** *For every integer $M \geq 64$ and $2 \leq r \leq M + 2$, put $\ell = 1 + \log((M+2)/r)$. Then*

$$2^{-30}\ell \leq \Gamma_{\mathrm{stat}}(M, r) \leq \Gamma_{\mathrm{all}}(M, r) \leq 2^{20}\ell. \tag{3.1}$$

The upper uses only exact counts and the suffix. The lower grants both boundary windows. All experiments use the supplied certificate $1/M$.

The proof in Section C checks the complete finite domain of the matched-observation bound. It tests the clean risk of the same map before selecting an attacked budget; it is not a logarithmic risk lower at each budget. Linear widths have constant factor; for fixed $0 < \gamma < 1$, $r = \max\{2, \lceil M^\gamma \rceil\}$ has order $\log M$, as does $r = 2$. The constants are conservative, not a claim of sharp practical thresholds.

The next sequence statements are consequences at the order level. Their original separate proofs are retained in Proposition B.4, including the stronger fixed-fraction lower constant.

Theorem 3.2 (Exact raw-window criterion)**.** *For every prescribed sequence of integers $2 \leq r_M \leq M + 2$,*

$$C_{\mathrm{stat}}(r_\bullet) < \infty \iff C_{\mathrm{all}}(r_\bullet) < \infty \iff \liminf_{M\to\infty} \frac{r_M}{M} > 0.$$

No monotonicity of r_M is required. The factor may depend on the prescribed sequence.

Necessity already uses stationary witnesses. Sufficiency includes unsupported starts. A finite initial set of horizons is harmless because $g_M(k) \geq M^{-1/2}$. The conclusion is about one budget-blind family, not a separate minimax estimator chosen for each k.

Theorem 3.3 (Price of a fixed fraction)**.** *Fix $0 < \alpha \leq 1/4096$ and $r_M = \max\{2, \lceil \alpha M \rceil\}$. Abbreviating the corresponding factors by $C_{\mathrm{stat}}(\alpha)$ and $C_{\mathrm{all}}(\alpha)$,*

$$2^{-26} \log \frac{1}{32\alpha} \leq C_{\mathrm{stat}}(\alpha) \leq C_{\mathrm{all}}(\alpha) \leq 2^{19}\left(1 + \log \frac{2}{\alpha}\right). \tag{3.2}$$

The upper requires only the histogram and suffix; the lower permits both boundary windows.

Thus the optimal uniform factor has order $\log(1/\alpha)$ as $\alpha \downarrow 0$. The upper is uniform in every M, k for each fixed α. The lower first sends $M \to \infty$ with α fixed and with a proposed finite factor fixed. This fixed-fraction proof takes limits in the stated order. The finite characterization above requires no limit in either parameter; neither result is a logarithmic risk lower bound at every budget. The constants are explicit but not optimized. Proposition B.4 proves Theorems 3.2 and 3.3 from a finite upper bound and a finite matched-observation lower bound.

3.1 An explicit estimator

Set

$$m = \min\{r-2, \lfloor M/2 \rfloor\}, \qquad N = M - m, \qquad L = 1 + \log \frac{M}{m+1}. \tag{3.3}$$

The last $m+2$ symbols reveal exactly the m departures indexed $M-m, \dots, M-1$ and the terminal pair. Let T'_u count visits to u at those departures and put $O'_u = J'_u - T'_u$. Then O'_u is the count on indices $0, \dots, N-1$; it is a direct old-prefix count even though it is computed by subtraction. At the selected row, predict designated probability

$$\widehat{q} = \begin{cases} C'/J', & J' > 0 \text{ and } 512O' \geq J', \\ 0, & \text{otherwise.} \end{cases} \tag{3.4}$$

The complementary symbol receives probability $1 - \widehat{q}$. This map does not use k or β. When $r = 2$, $m = 0$ and $O' = J'$; the terminal pair is not a departure. Using $r-1$ in place of $r-2$ would overcount the available information.

The finite upper bound at the main certificate, proved in Proposition A.9, is

$$R \leq \min\left\{1, \frac{131072}{\sqrt{M}} + (8448L + 147744)\frac{k}{M}\right\} \leq 287264 L g_M(k). \tag{3.5}$$

If r_M/M is eventually bounded below, then L is eventually bounded. For fixed α as in Theorem 3.3, $m + 1 \geq \alpha M/2$, including the horizons at which $r_M = 2$. This proves the two stated upper conclusions without a normalization loss.

4 Why positions enter the proof

Older visits and a truncated harmonic sum. The gate is not an estimate of the budget. It requires a positive fraction of the selected row's visits to precede the observed suffix, not merely one old occurrence. This proportional balance makes gate-switch errors chargeable to the edit budget; one forged old occurrence alone must not unlock a large corrupted suffix count. For a clean d-state homogeneous chain and any start, the relevant occupation inequality is

$$\mathbb{E}\frac{\mathbf{1}\{O \geq cJ\}}{J+1} \leq (3d^2 + 2d^3)\frac{1 + \log(M/(m+1))}{M}, \qquad c > 0. \tag{4.1}$$

Here J is the number of departures from the actual terminal state and O counts those before time $M - m$. Lemma A.7 proves this by an exact first-hit convolution and a reciprocal-return identity. In this analytic bound the event can be enlarged to the existence of an old visit. That enlargement removes short backward-time terms; the surviving harmonic range produces the logarithm. The implemented gate still requires proportional balance. Generic finite-state first-hit pointmass estimates have precedents in Norris et al. [11]; the required gated convolution is proved explicitly.

A strict k-edit outcome changes each relevant count by at most $6k$. A deterministic true-gate/false-gate split therefore reduces the error to a clean empirical comparator, the reciprocal count in (4.1), a weighted imbalance term, and two small-occupation events. The all-state certificate controls the imbalance even from unsupported starts. No conditional independence of departures given the random terminal row or its occupation is assumed. The supplement's row-stack argument instead starts from independent row-wise randomness and then bounds the selected empirical deviation.

Mixture domination and legal observation matching. The lower bound uses stationary alternatives whose designated exit probabilities are

$$(q_{00}, q_{01}, q_{10}, q_{11}) = (a, 1/2, 1/2, t), \qquad \Pi_a = \frac{(t/a, 2t, 2t, 1)}{1 + t/a + 4t}.$$

An integral over a creates the logarithmic term. Conditioning on a fixed departure count, one counts marked excursions and holding slots; the resulting occupation-mixture bound controls each proposed output mass. An interior loop relocation preserves the alternative's triple histogram and both of its observed windows, while making it close to a simple two-run null word. These are the alternative's windows, not windows of the unedited null input: changing an observed input-window symbol is legal but must be charged to the budget.

The attack normalizes only nonzero proposal masses, samples once, performs the relocation, and checks the actual Hamming cost and terminal pair. If any check fails, it returns its input unchanged. It is therefore a common source-independent kernel satisfying the budget on every outcome, not only on a high-probability event. In the risk comparison, nonnegative loss is discarded before proposal-mass domination; a signed triangle inequality is used only afterwards. Lemmas B.1 and B.2 and Proposition B.3 supply these steps.

For sublinear widths along any subsequence, a budget between the window and horizon scales makes the clean risk negligible relative to the alternative's exit scale while the mixture logarithm diverges. For fixed α, a budget proportional to the window yields $\log(1/\alpha)$. The alternatives may depend on this tested budget, but their clean risk is bounded using the same estimator at the same M and supplied $1/M$. This constrained-risk logic is related to general risk-transfer arguments such as Duchi and Ruan [4]; the dependent-word construction and legal observation matching are additional obligations.

5 One computed positional bit

Now change the access model. The decoder receives exact triple counts and the protected pair, optionally also the initial pair. An encoder may scan the entire already corrupted word and send one symbol in a fixed finite alphabet $\mathcal{A}$. It knows M but not k, β, or the source. The message is exact and fault-free after construction. There are no timing, framing, variable-length, or other side channels, and no shared source-dependent information.

Corollary 5.1 (Minimum supplemental alphabet)**.** *In this scan-and-message model, the minimum $|\mathcal{A}|$ for a finite uniform factor relative to $g_M(k)$ at supplied $1/M$ is two. This holds for stationary or arbitrary initialization, with or without the initial pair supplied.*

For the upper, put $n = \lfloor M/2 \rfloor$, select $u = Y_M$, and send

$$B = \mathbf{1}\{512J_1' \geq J'\}, \qquad J_1' = \sum_{i=0}^{n-1} \mathbf{1}\{(Z_i, Z_{i+1}) = u\}. \tag{5.1}$$

The decoder outputs C'/J' if $J' > 0$ and $B = 1$, and zero otherwise. Equality is accepted. If $J' = 0$, the bit equals one, but the explicit positive-denominator test enforces the fallback. Proposition A.9 gives all-start risk at most $\min\{1, 2^{17}M^{-1/2} + 2^{23}k/M\}$. A one-symbol alphabet supplies no information. Theorem 3.2 with $r_M = 2$ rules it out even when both endpoint pairs are present. This proves minimality without another lower-bound construction.

A finite word illustration, not a risk experiment. For $M = 65$, the two words in Table 1 have the same initial pair 01, terminal pair 10, and triple counts, in lexicographic order,

$$(H_{000}, H_{001}, H_{010}, H_{011}, H_{100}, H_{101}, H_{110}, H_{111}) = (0, 1, 1, 1, 1, 0, 1, 60).$$

At row 10, $J' = C' = 1$. The single departure from that row lies in different halves, so the computed gate differs although $\mathcal{T}_2$ is identical. The example verifies a combinatorial collision only: it gives no source probabilities, statistical risk comparison, or lower-bound proof.

Table 1: An exact 67-symbol collision. The notation 1^{62} means 62 consecutive ones; the first half contains departures 0 through 31.

Word	J'_1	B	Designated prediction
$0100\,1^{62}\,0$	1	1	1
$0\,1^{62}\,0010$	0	0	0

This is a worst-case supplemental alphabet result, not an entropy or mutual-information statement. The baseline exact counts are uncharged inputs. One computed bit is not one observed raw coordinate and does not remove the whole-word scan. A counter-based implementation is a natural baseline: maintain the eight triple counts and four first-half row counts, retain the endpoint, and select its row after the scan. This uses $O(\log M)$ counter workspace, $O(M)$ counter updates, and at most $O(M \log M)$ naive bit operations. No neural mechanism is needed for this statistical rate guarantee.

6 Relation to prior work and scope

Clean positional prediction. Han et al. [6, model (1), Theorem 24, and Section VI.A] study stationary prediction at the actual terminal state, including suffix-based estimators and higher-order KL analysis. Theorem 24 gives the binary order-two KL result; the squared-TV result in Section VI.A concerns first-order chains with fixed state count at least three. These are direct antecedents, not independent-query results. Their clean full-trajectory experiment differs from the strict-edit and restricted-observation comparison here. Losses must also be distinguished: a squared-TV upper bound implies an unsquared-TV upper bound by Jensen, but a squared-TV lower bound does not yield its square root as an unsquared-TV lower bound. The spectral-gap-restricted results in that work use an assumption different from the supplied all-state minorization (2.1).

Unknown contamination. Budget adaptation is not new in itself. Jain et al. [7] select among candidate estimates with monotone radii under a simultaneous-validity premise: every valid-index ball must contain the target. Separate expected-risk bounds do not establish that premise. Their selection lemma is pointwise, however, so the random terminal target is not by itself an obstruction: it could be applied realization by realization if all candidate predictions share the protected query and simultaneous accuracy is proved. Bhatt et al. [1] explicitly construct confidence-based adaptation for a mean under strict retrospective replacements of IID observations. Chen et al. [2] give robust testing and estimation under IID Huber mixtures, while Liu and Gao [10] exhibit an adaptation obstruction involving both smoothness and contamination. None of these comparisons licenses a general claim that unknown contamination alone prevents adaptation.

Dependence and count-based inference. Lecestre [8] estimate marginal or invariant distributions under dependence; their contamination corollary imposes independence conditions on the replacement channel. Leskelä and Dreveton [9] estimate transition matrices from independent parallel trajectories with arbitrarily corrupted whole trajectories. These are not the one-word terminal-target experiment. They also show why the present result should not be summarized as a universal failure of counts under corruption.

Clean transition-count factorization and Markov exchangeability remain valid. The report of Diaconis and Freedman [3] and the finite-stack discussion of Halberstam and Hutchcroft [5] provide relevant background on count-compatible paths. For a clean order-two word, likelihood is its initial-pair mass times transition probabilities raised to triple counts. After a position-sensitive kernel K, an observed word's probability instead sums these clean probabilities weighted by $K(z \mid x)$. Clean factorization alone does not establish sufficiency in this altered experiment. The specific observation map, not an unrestricted claim about all compact summaries, is essential.

Limits of the conclusions. The lower bounds concern uniform unknown-budget comparison at the critical supplied certificate, with exact counts and fixed boundary windows. They are not full-history impossibility results, every-budget logarithmic minimax penalties, claims about arbitrary interior statistics, or noisy-bit guarantees. The positive results include arbitrary starts; the lower witnesses are stationary. The constants are deliberately conservative. The word example is not empirical validation, and no learned-predictor experiment is asserted.

Computational realizations may be studied separately, but resource distinctions do not change these statistical statements. A whole-word counter baseline must be compared on the same input and arithmetic model. A finite-precision network realization would additionally need to account for width, internal precision, scanning, storage, and decoder cost. Neither existence of such a realization nor a depth bound would establish learnability, optimization success, Transformer superiority, or a general architecture lower bound. The present manuscript makes none of those claims.

A Finite-state estimates and upper bounds

Throughout, a departure count excludes the terminal state. For a chain $(Y_i)_{i=0}^{M}$, write

$$J_u = \sum_{i=0}^{M-1} \mathbf{1}\{Y_i = u\}, \qquad J = J_{Y_M}.$$

For the binary order-two experiment, $Y_i = (X_i, X_{i+1})$, $X_0, \dots, X_{M+1}$ contains M departures, and $q_u = \theta_u(d(u))$ with $d(a,b) = 1-a$. Let C_u count the corresponding designated successors and put $C = C_{Y_M}$ and $q = q_{Y_M}$. Primed counts are computed on the corrupted word. Every bound below permits an arbitrary initial law. The all-state certificate, when invoked, is $P^2(u,v) \geq \beta\Pi(v)$ for every pair of states, with $\beta > 0$. The final pair is protected, so clean and corrupted counts always select the same actual query row.

A.1 Universal occupation and clean prediction

Lemma A.1 (Actual-terminal occupation)**.** *For any homogeneous chain on d states, any initial law, any $M \geq 0$, and any integer $w \geq 1$,*

$$\Pr(J_{Y_M} < w) \leq \min\left\{1, \frac{d^2 w}{M+1}\right\}.$$

No irreducibility or stationary-start assumption is needed.

Proof. For each t set $J(t) = \sum_{i<t} \mathbf{1}\{Y_i = Y_t\}$, $b_t(v) = \Pr_v(J(t) < w)$, and $B_t = \max_v b_t(v)$. If $t \geq \ell$, the full count dominates the count in the final ℓ-step suffix. Conditioning on the suffix's initial state and using homogeneity gives $B_t \leq B_\ell$. On every path, each state can occur at most w times with fewer than w earlier occurrences. Hence $\sum_{t=0}^{M} \mathbf{1}\{J(t) < w\} \leq dw$. Consequently

$$(M+1)B_M \leq \sum_{t=0}^{M} B_t \leq \sum_{v} \sum_{t=0}^{M} b_t(v) \leq d^2 w.$$

Average over the initial law and use the probability cap. □

Lemma A.2 (Binary clean empirical comparison)**.** *For every homogeneous binary order-two source, every initial law, and $M \geq 64$,*

$$\mathbb{E}\left[\mathbf{1}\{J > 0\}\left|\frac{C}{J} - q\right|\right] \leq \frac{512}{\sqrt{M+1}}.$$

The expression inside the indicator is evaluated only when $J > 0$.

Proof. For each of the four rows u, generate an independent stack of $2M$ Bernoulli(q_u) variables $\xi_{u,j}$, independently of the initial pair. At the jth departure from u, use $\xi_{u,j}$ to choose whether the successor is $d(u)$. For every fixed finite path, the used stack prefixes have exactly the product of its transition probabilities; thus this realizes the original law, including boundary probabilities. Write $S_{u,j} = \sum_{i=1}^{j}(\xi_{u,i} - q_u)$. Pathwise, $C - Jq = S_{Y_M,J}$ on $J > 0$. The stacks are not asserted independent after conditioning on J, Y_M, or any selection event.

For a centered variable in an interval of length one, its log moment generating function has second derivative equal to a tilted variance, which is at most $1/4$. Integrating twice gives $\mathbb{E}e^{\lambda V} \leq e^{\lambda^2/8}$.

Thus $e^{\lambda S_{u,j} - j\lambda^2/8}$ is a nonnegative supermartingale. Stopping at its first level crossing before the deterministic time $2v$ gives, for $x > 0$,

$$\Pr\left(\max_{j \le 2v} S_{u,j} \ge x\right) \le e^{-\lambda x + 2v\lambda^2/8} \le e^{-x^2/v}, \qquad \lambda = 2x/v.$$

The stopping inequality here is finite: the expectation of the stopped process is at most its initial value by summing conditional expected increments through time $2v$. Apply the same argument to $-S$ and take a union over rows. For $Z_v = \max_u \max_{j \le 2v} |S_{u,j}|$,

$$\Pr(Z_v \ge x) \le 8e^{-x^2/v}. \tag{A.1}$$

For dyadic integers $v \le M$, let $E_v = \{v \le J < 2v\}$ and $p_v = \min\{1, 32v/(M+1)\}$. Lemma A.1 gives $\Pr(E_v) \le p_v$. Without any independence assumption, tail integration and $a = \sqrt{v \log(8/p_v)}$ yield

$$\mathbb{E}[Z_v \mathbf{1}_{E_v}] \le \int_0^\infty \min\{p_v, 8e^{-x^2/v}\}\, dx \le p_v a + \frac{p_v v}{2a} \le 2p_v\sqrt{v \log(8/p_v)}.$$

For the middle inequality, bound the Gaussian tail above a by inserting $x/a \ge 1$ and integrating. This layer contributes at most $2p_v\sqrt{\log(8/p_v)}/\sqrt{v}$ to the desired risk. Put $N = M + 1$. Summing layers with $v \ge N/32$ costs at most

$$\frac{2\sqrt{32 \log 8}}{1 - 2^{-1/2}} N^{-1/2} < 56N^{-1/2}.$$

If v_0 is the largest dyadic integer below $N/32$, then $N/64 \le v_0 < N/32$ and $\log(N/(4v_0 2^{-j})) \le 3(j+1)$. The remaining layers cost at most

$$\frac{64\sqrt{3}}{\sqrt{32N}} \sum_{j \ge 0} 2^{-j/2}\sqrt{j+1} \le \frac{64\sqrt{3}}{\sqrt{32N}}(1 - 2^{-1/2})^{-2} < 236N^{-1/2}.$$

Only a numerical series was extended; all stack maxima use at most $2M$ variables. The layers partition $1 \le J \le M$, including the last layer when M is dyadic. The safe constant 512 follows. □

A.2 Return cycles and binary hitting scales

Lemma A.3 (Finite-state return identity)**.** *In a finite irreducible chain with stationary law p, the first positive return T to u has mean $\mathbb{E}_u T = 1/p_u$. If $H_u = \max_v \mathbb{E}_v D_u$, where D_u is the ordinary hitting time, zero at u, then $1/p_u \le 1 + H_u$.*

Proof. Irreducibility and finiteness give a fixed block length and a positive minimum probability of hitting u within a block, uniformly over starting states. Repeated conditioning gives geometric tails for hitting and return times, so the following expectations are finite. Let $\nu(v) = \mathbb{E}_u \sum_{j=0}^{T-1} \mathbf{1}\{Y_j = v\}$. First-step conditioning inside this sum gives

$$(\nu P)(v) = \mathbb{E}_u \sum_{j=1}^{T} \mathbf{1}\{Y_j = v\} = \nu(v),$$

since $Y_0 = Y_T = u$. A finite irreducible chain has a unique stationary probability: for completeness, if η is another one, the ratios $\eta(v)/p_v$ are harmonic for the reversed transition matrix; a maximal ratio propagates along every positive reversed edge and hence is constant. Thus $\nu = (\mathbb{E}_u T)p$. Exactly one visit to u occurs before T, so $\nu(u) = 1$, proving the identity. A positive return is one step followed by an ordinary hit, proving $\mathbb{E}_u T \le 1 + H_u$. □

Lemma A.4 (Support and weighted hitting scale)**.** *A binary order-two source with a positive all-state certificate has either full stationary support or a singleton absorbing diagonal support. In the full-support case, with $q = (a, b, e, \tau)$ and $\overline{H}_u = \max\{H_u, 1/\Pi(u)\}$,*

$$\sum_u \Pi(u) q_u \overline{H}_u \le \frac{40}{\sqrt{\beta}}. \tag{A.2}$$

Proof. Stationarity of consecutive pair marginals gives $\Pi = (y, x, x, w)$, and stationary flow into each diagonal gives

$$ya = xe, \qquad w\tau = xb. \tag{A.3}$$

If $x > 0$, the certificate entries $P^2(10, 01) = ea \ge \beta x$ and $P^2(01, 10) = b\tau \ge \beta x$ force $a, b, e, \tau > 0$. The flow identities then imply $y, w > 0$. If $x = 0$, the stationary law is on diagonals; each supported diagonal is absorbing by stationarity. Two absorbing supported diagonals contradict the positive certificate. This proves the classification. Summing a certificate row gives $\beta \le 1$. Full support and the certificate imply irreducibility.

Assume full support and put $\gamma = (a, 1, 1, \tau) \ge q$. The two certificate entries just used, together with (A.3), also give

$$e^2 \ge \beta y, \qquad b^2 \ge \beta w. \tag{A.4}$$

Here, for example, $ea \ge \beta x = \beta ya/e$ and $a, e > 0$. For target 00, write h_{01}, h_{10}, h_{11} for the mean hit times. The first-step equations are

$$h_{01} = 1 + (1 - b)h_{10} + bh_{11}, \quad h_{10} = 1 + (1 - e)h_{01}, \quad h_{11} = \tau^{-1} + h_{10}.$$

They give

$$h_{01} = \frac{b + 2\tau}{e\tau}, \quad h_{10} = h_{01} - \frac{b}{\tau} - 1, \quad h_{11} = h_{01} + \frac{1 - b}{\tau} - 1,$$

and therefore $H_{00} \le b/(e\tau) + 2/e + 1/\tau + 1$. For target 01, the three hit times from $00, 10, 11$ are $1/a$, $1 + e/a$, and $1/\tau + 1 + e/a$, so $H_{01} \le 2/a + 1/\tau + 2$. Exchanging symbols gives the other two bounds:

$$H_{10} \le 1/a + 2/\tau + 2, \qquad H_{11} \le e/(ba) + 2/b + 1/a + 1.$$

The substitution exchanges 00 with 11, 01 with 10, and (a, b, e, τ) with (τ, e, b, a); it does not assume $b = e$. Using (A.3) and (A.4),

$$\sqrt{\beta}\, yaH_{00} \le \sqrt{\beta}\,(w + 2x + ew/b + xe) \le \sqrt{\beta}\,(w + 3x) + \sqrt{w} \le 5,$$
$$\sqrt{\beta}\, xH_{01} \le 2\sqrt{\beta}\, y/e + \sqrt{\beta}\, w/b + 2\sqrt{\beta}\, x \le 2\sqrt{y} + \sqrt{w} + 2x \le 5.$$

The symbol-exchanged inequalities give the same bound for $10, 11$. Lemma A.3 implies $\overline{H}_u \le H_u + 1$; hence $\Pi(u)\gamma_u \overline{H}_u \le 6/\sqrt{\beta} \le 10/\sqrt{\beta}$. Sum over four states and use $q \le \gamma$. □

A.3 First hits, endpoint probabilities, and joint tails

Lemma A.5 (Target endpoint and occupation bounds)**.** *For any d-state chain, possibly reducible, and $v \ne u$,*

$$\Pr_v(D_u = t) \le \frac{d}{t + 1}, \qquad t \ge 1. \tag{A.5}$$

For an irreducible chain, fix a target with stationary mass $p > 0$, maximum ordinary hitting mean H, and $h = \max\{H, 1/p\}$. For every initial state i and every integer $\ell \geq 1$, with $J_u(\ell) = \sum_{j=0}^{\ell-1} \mathbf{1}\{Y_j = u\}$,

$$\Pr_i(Y_\ell = u) \leq 4dp(1 + h/\ell), \tag{A.6}$$

$$\Pr_i(J_u(\ell) < \ell p/16) \leq 2e^{-\ell/(8h)}, \tag{A.7}$$

$$\Pr_i(Y_\ell = u, J_u(\ell) > 4\ell p + 1) \leq 16dp(1 + h/\ell)e^{-\ell/(2h)}. \tag{A.8}$$

Proof. Reciprocal return coefficients and first hits. For any finite stochastic matrix P, let T_j be the jth positive return to u, allowing defective returns, and let $F(z) = \mathbb{E}_u[z^{T_1}; T_1 < \infty]$. Strong Markov at returns gives $\Pr_u(T_j = s) = [z^s]F(z)^j$. All strong-Markov uses here can be obtained by conditioning on each possible stopping time and state and multiplying the subsequent transition probabilities; no conditional independent-sample claim is involved. Let P_{-u} delete row and column u. Writing P in target/rest block form, direct summation over the excursion before the first return gives $F(z) = zP_{uu} + z^2 P_{u,-u}(I - zP_{-u})^{-1}P_{-u,u}$. The block determinant identity therefore yields

$$1 - F(z) = \frac{\det(I - zP)}{\det(I - zP_{-u})}.$$

These are formal series, so no convergence issue arises. The identity $-\log\det(I - zA) = \sum_{s\geq 1} \operatorname{tr}(A^s)z^s/s$ follows by differentiating the determinant, expanding $(I - zA)^{-1}$, and integrating from zero. Consequently

$$\sum_{j\geq 1} \frac{\Pr_u(T_j = s)}{j} = \frac{\operatorname{tr}(P^s) - \operatorname{tr}(P^s_{-u})}{s} \leq \frac{d}{s}, \qquad s \geq 1. \tag{A.9}$$

Its left side is nonnegative; the upper bound uses nonnegative entries of P^s_{-u} and diagonal entries of P^s at most one. Now change only row u to send u deterministically to $v \neq u$. In this auxiliary stochastic chain, the first positive return has law $1 + D_u$ under the original start v. Retain only the first-return term of (A.9) to prove (A.5). The auxiliary chain need not be an order-two source.

A target-specific endpoint bound. Return now to the irreducible chain. Let $g(v) = \mathbb{E}_v D_u$. First-step equations and Lemma A.3 give $(Pg - g)(v) = -1 + \mathbf{1}\{v = u\}/p$. Telescoping expectations shows

$$\mathbb{E}_v J_u(s) = p\{s + \mathbb{E}_v g(Y_s) - g(v)\} \leq p(s + H). \tag{A.10}$$

A loop-rotation identity is

$$sP^s(u, u) = \sum_v \mathbb{E}_v[J_u(s)\mathbf{1}\{Y_s = v\}] \leq dp(s + H).$$

Indeed, for each $j < s$, the corresponding sum is $\sum_v P^j(v, u)P^{s-j}(u, v) = P^s(u, u)$. For $v \neq u$, split the first-hit convolution at $t = \lfloor s/2 \rfloor$. For early hits the return factor is at most $dp(1 + 2H/s)$. For late hits (A.5) is at most $2d/s$, and (A.10) from u bounds the sum of return factors by $p(\lceil s/2 \rceil + H)$. The resulting bound is at most $dp(2 + 2/s + 4H/s) \leq 4dp(1 + H/s)$. It also holds for $v = u$; the early sum may be empty at $s = 1$. This proves the slightly stronger version of (A.6) with H instead of h.

Return moments and positive transforms. The family consisting of ordinary hitting times from nontarget states and the positive return from u is closed under conditioning on survival and the current state. Its means are at most h. For each member V and integer $r \geq 1$,

$$\mathbb{E}V^r \leq r!h^{r-1}\mathbb{E}V. \tag{A.11}$$

To prove this by induction, use $V^r \le r\sum_{j=0}^{V-1}(V-j)^{r-1}$, condition each summand on survival through time j, and apply the induction hypothesis and the uniform mean bound h to the residual law. The sum of the survival probabilities is $\mathbb{E}V$. With $\lambda = 1/(4h)$, expansion in nonnegative powers gives

$$\mathbb{E}_i e^{\lambda D_u} \le \frac{4}{3}, \qquad \mathbb{E}_u e^{\lambda T_1} \le e^{2\lambda/p}, \qquad f := \mathbb{E}_u e^{-\lambda T_1} \le e^{-3\lambda/(4p)}. \tag{A.12}$$

For the last inequality use $e^{-x} \le 1 - x + x^2/2$ and $\mathbb{E}T_1^2 \le 2h/p$; the first two follow from the geometric-series bound on the moments. If $K = \lceil \ell p/16 \rceil$, fewer than K visits in $0, \dots, \ell - 1$ means $D_u + T_1 + \cdots + T_{K-1} \ge \ell$. These are fresh independent return cycles after the first hit, before any horizon restriction. Exponential Markov and (A.12) bound this probability by $(4/3)e^{-\lambda\ell + 2\lambda(K-1)/p} \le (4/3)e^{-7\lambda\ell/8} \le 2e^{-\ell/(8h)}$. This includes $K = 1$ and proves (A.7).

A joint tail by an exact change of path measure. Let $h_i = \mathbb{E}_i e^{-\lambda D_u}$, with $h_u = 1$. The inequality $e^{-x} \ge 1 - x$ gives $h_i \ge 3/4$ and $f \ge 3/4$. Define

$$Q_{ij} = e^{-\lambda} P_{ij} h_j / h_i \quad (i \ne u), \qquad Q_{uj} = e^{-\lambda} P_{uj} h_j / f.$$

The hitting and return equations show that every row sums to one. This auxiliary matrix preserves directed support, so is irreducible. Its ordinary hitting law is the original law tilted by $e^{-\lambda D_u}/h_i$, whence $H_Q \le 2H$. Its positive return law is tilted by $e^{-\lambda T_1}/f$, and

$$\frac{1}{2p} \le \mathbb{E}_Q T_1 \le \frac{2}{p}.$$

For the lower bound, use $\mathbb{E}[T_1 e^{-\lambda T_1}] \ge 1/p - \lambda(2h/p) = 1/(2p)$ and $f \le 1$; the upper bound uses $f \ge 3/4$. Kac for Q gives $p_Q \le 2p$ and $\max(H_Q, 1/p_Q) \le 2h$. The endpoint bound already proved yields $Q^\ell(i,u) \le 16dp(1 + h/\ell)$. For every length-$\ell$ path from i to u, telescoping the factors gives

$$Q(\text{path}) = P(\text{path}) e^{-\lambda\ell} f^{-J_u(\ell)} / h_i.$$

The count is exactly the number of target departures. Thus on $J_u(\ell) > 4\ell p + 1$, summing the reverse likelihood relation gives

$$\Pr_i(Y_\ell = u, J_u(\ell) > 4\ell p + 1) \le h_i e^{\lambda\ell} f^{4\ell p} Q^\ell(i,u) \le 16dp(1 + h/\ell) e^{-\ell/(2h)}.$$

This proves the joint bound with its stationary-mass factor; no product of an unconditional upper tail and an endpoint probability was taken. □

A.4 Weighted imbalance, including unsupported starts

Lemma A.6 (Half-window imbalance)**.** *For $M \ge 64$, put $n = \lfloor M/2 \rfloor$ and $J_{1,u} = \sum_{i=0}^{n-1} \mathbf{1}\{Y_i = u\}$. Every binary source with a positive all-state certificate, under every initial law, satisfies*

$$\mathbb{E}\left[q_{Y_M} \mathbf{1}\{J > 0, J_{1,Y_M} < J/256\}\right] \le \frac{65536}{M\sqrt{\beta}}.$$

Proof. We first prove a general irreducible-chain statement. Let $m = M - n$, $J_{2,u} = \sum_{i=n}^{M-1} \mathbf{1}\{Y_i = u\}$, and assign arbitrary weights $0 \le q_u \le 1$. Write $W_u = p_u q_u \overline{H}_u$, where p_u is stationary mass. For every $0 < c \le 1/256$ we claim

$$\mathbb{E}\left[q_{Y_M} \mathbf{1}\{J_{1,Y_M} < cJ\}\right] \le \frac{288d}{M} \sum_u W_u. \tag{A.13}$$

The strict imbalance event is empty at $J = 0$. Fix u with $np_u \geq 1$. If $J_{1,u} \geq np_u/16$ and $J_{2,u} \leq 4mp_u + 1$, then $m \leq 2n$ gives $J_{1,u}/(J_{1,u} + J_{2,u}) \geq 1/145 > 1/256$. Thus imbalance requires a first-window lower tail or a second-window upper tail. For the first event, condition on the whole history through Y_n, use the uniform future endpoint bound (A.6), and then the first-window lower tail (A.7). For the second event condition only on Y_n and use the joint bound (A.8). Since $m \geq n$,

$$\Pr_i(Y_M = u,\ J_{1,u} < cJ_u) \leq 24dp_u(1 + \overline{H}_u/n)e^{-n/(8\overline{H}_u)}.$$

After multiplication by q_u, this is $(24dW_u/n)(x+1)e^{-x/8}$, where $x = n/\overline{H}_u$. Since the maximum of $(x+1)e^{-x/8}$ is $8e^{-7/8} < 4$, it is at most $96dW_u/n$. If $np_u < 1$, use only the endpoint bound at time M. As $W_u \geq q_u$ and $p_u q_u < q_u/n$,

$$q_u \Pr_i(Y_M = u) \leq 4d(p_u q_u + W_u/M) \leq 8dW_u/n.$$

Sum over u, average over initial states, and use $n \geq M/3$ to prove (A.13). For full-support binary sources, Lemma A.4 now gives $288 \cdot 4 \cdot 40/(M\sqrt{\beta}) = 46080/(M\sqrt{\beta})$.

It remains to treat singleton stationary support, without discarding unsupported actual initial states. Suppose 00 is absorbing, so $a = 0$, and denote the other states by $A = 01$, $B = 10$, $C = 11$. An original path ending in this active set has never hit 00. Reflect the killing edge $B \to 00$ to $B \to A$, producing the auxiliary three-state chain

$$R(A,B) = 1 - b, \quad R(A,C) = b, \quad R(B,A) = 1, \quad R(C,B) = \tau, \quad R(C,C) = 1 - \tau.$$

Every original active path has probability at most its reflected-path probability: the only changed factor is $1 - e \leq 1$ on $B \to A$. The endpoint and all active-state counts are unchanged. Thus any nonnegative functional of these paths is dominated without conditioning on survival. For $b, \tau > 0$ this chain is irreducible, with

$$Z = 2\tau + b, \quad p_A = p_B = \tau/Z, \quad p_C = b/Z, \qquad p_A b\overline{H}_A \leq 1, \quad p_C \tau \overline{H}_C \leq 1. \tag{A.14}$$

Indeed $H_A = 1 + 1/\tau$ and $H_C = 2/b$ by first-step equations; $\overline{H}_A \leq 2 + 1/\tau$ and $\overline{H}_C \leq 1 + 2/b$ give the last inequalities since $b, \tau \leq 1$. Apply (A.13) with weights $(b, 0, \tau)$: the A, C contribution is at most $288 \cdot 3 \cdot 2/M = 1728/M$. If $b = 0$ or $\tau = 0$, approximate these two parameters by positive ones in the reflected chain. At fixed finite horizon, path probabilities and the weights are continuous, and the bound is uniform, so the same conclusion follows by taking a limit. No minorization-preserving approximation of the original source is being asserted. The remaining target B has weight e, and for an active start

$$e \Pr(Y_M = B) = \Pr(D_{00} = M + 1) \leq \frac{4}{M+2}, \tag{A.15}$$

by (A.5): the sole entry edge to 00 is $B \to 00$. Drop its imbalance event. An initial 00 contributes zero; mixtures follow by linearity. The total is at most $1732/M$, below the stated bound because $\beta \leq 1$. Exchanging symbols handles absorbing 11 with the same fixed designated-successor rule. □

A.5 Reciprocal occupation with an old visit

Lemma A.7 (Old-visit reciprocal count)**.** *For any homogeneous d-state chain and initial law, let $M \geq 2$, $0 \leq m \leq \lfloor M/2 \rfloor$, and*

$$O_u = \sum_{i=0}^{M-m-1} \mathbf{1}\{Y_i = u\}, \qquad O = O_{Y_M}.$$

For every $c > 0$,

$$\mathbb{E}\frac{\mathbf{1}\{O \geq cJ\}}{J+1} \leq \frac{1}{M}\left(3d^2 + 2d^3 \sum_{s=m+1}^{\lfloor (M-1)/2 \rfloor} \frac{1}{s}\right)$$
$$\leq \frac{3d^2 + 2d^3}{M}\left(1 + \log\frac{M}{m+1}\right),$$

where an empty sum is zero. In particular, the last numerator constant is 176 *for the pair chain. The statement includes $J = 0$ and $m = 0$.*

Proof. Starting at u, the event $\{Y_s = u, J_u(s) = j\}$ is exactly $\{T_j = s\}$ for $j \geq 1$: the initial visit is a departure, while the terminal visit is not. Equation (A.9), which allowed defective returns, gives

$$K_u(s) := \mathbb{E}_u \frac{\mathbf{1}\{Y_s = u\}}{J_u(s)+1} = \sum_{j \geq 1} \frac{\Pr_u(T_j = s)}{j+1} \leq \frac{d}{s}.$$

Fix an initial state $v \neq u$, and put $a_v(t) = \Pr_v(D_u = t)$. The event $Y_M = u, J_u = 0$ means $D_u = M$, so its contribution is at most $d/(M+1) \leq d/M$ by (A.5); it is impossible if $v = u$. On $J_u > 0, O_u \geq cJ_u$, integrality forces $O_u \geq 1$, hence the first hit is at most $M - m - 1$. Discarding the rest of the gate and using strong Markov gives the exact identity

$$\mathbb{E}_v \frac{\mathbf{1}\{Y_M = u, O_u \geq 1\}}{J_u + 1} = \sum_{t=1}^{M-m-1} a_v(t) K_u(M-t).$$

There is no target departure before t, so the full count is exactly the post-hit departure count, with no additional unit shift. For $t \leq M/2$, use $K_u(M-t) \leq 2d/M$ and $\sum_t a_v(t) \leq 1$; the contribution is at most $2d/M$. This also covers $v = u$, whose expression is just $K_u(M)$. For $t > M/2$, use $a_v(t) \leq 2d/M$ and write $s = M - t$; the contribution is at most

$$\frac{2d^2}{M} \sum_{s=m+1}^{\lfloor (M-1)/2 \rfloor} \frac{1}{s}.$$

Together with the zero-count term this is a per-target bound. Average over the start and sum over d terminal targets to obtain the first inequality. Finally, $\sum_{s=a}^{b} 1/s \leq 1/a + \log(b/a) \leq 1 + \log(M/a)$ for $b \leq M$, with the empty case automatic, gives the stated logarithmic bound. □

A.6 Protected edits and the two positional gates

Lemma A.8 (Pathwise gate reduction)**.** *Let D_u be a count of row u over a fixed set of departure indices, and at the protected row put $D = D_{Y_M}$. On the corrupted word use*

$$\widehat{q} = \begin{cases} C'/J', & J' > 0 \text{ and } 512D' \geq J', \\ 0, & \text{otherwise.} \end{cases}$$

For every allowed corruption kernel and every initial law,

$$R \leq \mathbb{E}\left[\mathbf{1}\{J > 0\}|C/J - q|\right] + 48k\mathbb{E}\frac{\mathbf{1}\{D \geq J/1024\}}{J+1} + \mathbb{E}[q\mathbf{1}\{J > 0, D < J/256\}]$$
$$+ \Pr(J < 6156k + 1) + \Pr(J < 3078k + 1). \tag{A.16}$$

In particular this applies to a first-half count or to the old-prefix count reconstructed by subtracting a suffix count from the total.

Proof. One edited symbol affects at most three departure triples; each changed triple changes the histogram's ℓ^1 norm by at most two. Pair-count windows affect at most two departures. Taking a union over the edits, including adjacent edits and indices straddling a split, gives the safe bounds

$$|J' - J|,\ |C' - C|,\ |D' - D| \le \delta := 6k.$$

When the old count is computed by subtraction, it is still exactly a direct count on each word, so the same bound applies without adding two separate perturbation bounds. Endpoint protection keeps the selected row fixed. Let $G' = \{J' > 0, 512D' \ge J'\}$, $H = \{D \ge J/1024\}$, and $B = \{J > 0, D < J/256\}$. On $G' \cap H$ with $J > 0$,

$$|C'/J' - q| \le |C/J - q| + \frac{8\delta}{J+1}.$$

For $\delta = 0$ the ratios coincide. For $\delta > 0$ and $J \ge 2\delta$, $J' \ge J/2$ and ratio subtraction gives a difference at most $4\delta/J \le 8\delta/(J+1)$. For $J < 2\delta$, use the unit bound on the difference of two probabilities. On $G' \cap H^c$, $D + \delta \ge D' \ge (J - \delta)/512$ whereas $D < J/1024$, so $J < 1026\delta = 6156k$. On $(G')^c \cap B^c$ with $J > 0$, if $J' = 0$ then $J \le \delta$; otherwise $D' < J'/512$ and $D \ge J/256$ give $J/256 - \delta < (J + \delta)/512$, hence $J < 513\delta = 3078k$. If the gate rejects on B, its loss is exactly q. The two displayed integer-threshold tails also cover $J = 0$. These cases give (A.16) pathwise and then in expectation, with $8\delta = 48k$. At $k = 0$ the false-status cases for $J > 0$ are impossible. Equality in the gate is accepted throughout. Since the comparison is pathwise, arbitrary permitted attack randomization does not change it. □

Proposition A.9 (Finite suffix and half-gate upper bounds)**.** *Let $M \ge 64$, $2 \le r \le M + 2$, and set*

$$m = \min\{r - 2, \lfloor M/2 \rfloor\}, \quad N = M - m, \quad L = 1 + \log \frac{M}{m+1}.$$

From the histogram obtain J'_u, C'_u, and from the last $m+2$ symbols obtain $T'_u = \sum_{i=M-m}^{M-1} \mathbf{1}\{Y'_i = u\}$. Use the gate in Lemma A.8 with $D'_u = O'_u = J'_u - T'_u$. This k, β-blind estimator has all-start risk

$$R \le \min\left\{1, \frac{65536}{\sqrt{M}} + \frac{65536}{M\sqrt{\beta}} + (8448L + 147744)\frac{k}{M}\right\}. \tag{A.17}$$

Alternatively, use $D'_u = J'_{1,u}$, the count over the first $\lfloor M/2 \rfloor$ departures. Its all-start risk is

$$R \le \min\left\{1, \frac{65536}{\sqrt{M}} + \frac{65536}{M\sqrt{\beta}} + 2^{23}\frac{k}{M}\right\}. \tag{A.18}$$

The latter estimator factors exactly through the histogram, protected pair, and the one bit $\mathbf{1}\{512J'_{1,Y_M} \ge J'_{Y_M}\}$.

Proof. The last $m + 2$ raw symbols reveal exactly m departure pairs and the terminal pair. Thus O'_u is precisely the count over indices $0, \dots, N - 1$, even though their raw order is not observed. In particular $r = 2$ means $m = 0$ and $O' = J'$; counting $r - 1$ suffix departures would incorrectly include the terminal pair. Since $N \ge \lceil M/2 \rceil$, the clean O is at least J_1. Lemma A.6 therefore bounds the bias in (A.16) by $65536/(M\sqrt{\beta})$. Lemma A.7, with $d = 4, c = 1/1024$, bounds its reciprocal term by $48k \cdot 176L/M = 8448Lk/M$. Lemma A.1 bounds the two tails together by

$$\frac{16(6156 + 3078)k + 32}{M+1} = \frac{147744k + 32}{M+1}.$$

Lemma A.2 and the remaining $32/(M+1)$ are bounded by $65536/\sqrt{M}$. The unit loss cap proves (A.17).

For the half gate we require a different reciprocal estimate, valid for any four-state chain and start:

$$\mathbb{E}\frac{\mathbf{1}\{J_1 \geq J/1024\}}{J+1} \leq \frac{130944}{M}. \tag{A.19}$$

To prove it, fix u and condition on the entire first-half history, whose terminal state is v and whose count of u is j. Let $m_2 = M - \lfloor M/2 \rfloor$ be the remaining length. Balance implies $J_{2,u} \leq 1023j$. In the suffix, the fixed-terminal event $Y_M = u, J_{2,u} \leq 1023j$ is a subset of the actual-terminal low-occupation event. Lemma A.1 gives conditional probability at most $16(1023j+1)/(m_2+1)$. Since $(J_u+1)^{-1} \leq (j+1)^{-1}$, the conditional contribution is at most $16368/(m_2+1)$, including $j=0$. Sum over four targets and use $m_2 + 1 \geq M/2$ to prove (A.19). This argument uses a deterministic prefix, not endpoint-conditioned independent samples; Lemma A.1 is valid at every suffix length. Apply Lemmas A.8, A.6, and A.2 again. The total corruption coefficient is

$$48 \cdot 130944 + 16(6156 + 3078) = 6433056 < 2^{23},$$

proving (A.18). The stated bit is exactly the gate decision on every corrupted word. If $J' = 0$ its value is one, but the decoder's separate $J' > 0$ test still returns zero. Thus neither this boundary case nor threshold equality requires another message. All edits occur before the exact message is formed; the bit channel itself is not being corrupted.

At $\beta = 1/M$, (A.17) gives directly, for $g_M(k) = \max\{M^{-1/2}, k/M\}$,

$$R \leq (131072 + 156192L)g_M(k) \leq 287264Lg_M(k) < 2^{19}Lg_M(k).$$

For $r = \max\{2, \lceil \alpha M \rceil\}$ with $0 < \alpha \leq 1$, $m+1 = \min\{r-1, \lfloor M/2 \rfloor + 1\} \geq \alpha M/2$. Indeed $r - 1 \geq \alpha M/2$ follows by separating $\alpha M \leq 2$ from $\alpha M > 2$. Hence $L \leq 1 + \log(2/\alpha)$. Likewise (A.18) becomes $\min\{1, 2^{17}/\sqrt{M} + 2^{23}k/M\}$ at the critical certificate. □

A.7 A count-only oracle when the budget is supplied

Proposition A.10 (Critical known-budget upper)**.** *At supplied certificate $\beta = 1/M$, a predictor using only the exact triple histogram, the protected pair, and the public budget k satisfies, for every source and initial law,*

$$R \leq 191488 \max\{M^{-1/2}, k/M\}, \qquad M \geq 64.$$

For $k \geq 1$ its designated probability is

$$\widehat{q}_k = \frac{(C' - 6k)_+}{J' + 6k + 2};$$

for $k = 0$ use $(C'+1)/(J'+2)$. This budget-dependent estimator is used only for the oracle comparison, not for the unknown-budget results.

Proof. A weighted occupation estimate. For an irreducible d-state chain with target masses p_u, let $h_u = \max\{H_u, 1/p_u\}$, $0 \leq q_u \leq 1$, and $W_u = p_u q_u h_u$. For every $A > 0$ and every initial law we first prove

$$\mathbb{E}\min\left\{q_{Y_M}, \frac{A}{J+2}\right\} \leq 384d^2\frac{A}{M} + \frac{192d}{M}\sum_u W_u. \tag{A.20}$$

Fix a target u. If $h_u > M$, the endpoint bound gives $q_u \Pr(Y_M = u) \le 4dp_uq_u(1+h_u/M) \le 8dW_u/M$. If $h_u \le M$, on $J_u \ge Mp_u/48$ the minimum is at most $48A/(Mp_u)$, and $\Pr(Y_M = u) \le 8dp_u$; this costs at most $384dA/M$. For the complementary event, put $n = \lfloor M/2 \rfloor$ and $m_2 = M - n$. Since $n \ge M/3$, $J_u < Mp_u/48$ implies $J_{1,u} < np_u/16$. Conditioning on the complete prefix through Y_n, use the uniform endpoint bound on the remaining m_2 transitions, and then the unconditional lower tail on the prefix. The weighted contribution is at most

$$8dp_uq_u(1 + 2h_u/M)e^{-M/(24h_u)} = \frac{8dW_u}{M}(x+2)e^{-x/24} \le \frac{192dW_u}{M}, \qquad x = M/h_u.$$

The last step uses $\sup_{x\ge 0}(x+2)e^{-x/24} = 24e^{-11/12} < 24$. Summing these two cases over targets proves (A.20). For full-support binary sources, Lemma A.4 yields

$$\mathbb{E}\min\left\{q_{Y_M}, \frac{A}{J+2}\right\} \le 6144\frac{A}{M} + \frac{30720}{M\sqrt{\beta}}. \tag{A.21}$$

For singleton support 00, the 00 contribution vanishes. Use the reflected chain from the proof of Lemma A.6. On paths ending active, original path weights are bounded above by reflected path weights, with exactly the same terminal state and departure counts. Thus the nonnegative minimum functional for targets A, C, with weights $(b, 0, \tau)$, is dominated pathwise in the same sum over paths. Equation (A.14) and (A.20), with $d = 3$, bound this part by $3456A/M + 1152/M$. The original target B contributes at most $e\Pr(Y_M = B) \le 4/(M+2)$ by (A.15). An initial 00 contributes zero. For boundary b or τ, approximate only the reflected chain by positive parameters: there are finitely many length-M paths, and each summand is continuous in probabilities and weights, including its minimum. The uniform bound therefore passes to the limit. Symbol exchange handles singleton 11. Since $\beta \le 1$, the larger bound (A.21) covers both singleton cases and arbitrary initial mixtures as well.

A pathwise lower envelope. For $k \ge 1$ put $B_k = 6k$, $D = J + 2$, $D' = J' + B_k + 2$, and $c' = (C' - B_k)_+$. The protected-edit perturbation bounds give

$$D' \ge D, \quad 0 \le c' \le C, \quad 0 \le C - c' \le 2B_k, \quad 0 \le D' - D \le 2B_k.$$

Thus $0 \le \widehat{q}_k \le C/(J+2)$, and

$$\frac{C}{D} - \frac{c'}{D'} = \frac{C - c'}{D} + \frac{c'}{D'}\frac{D' - D}{D} \le \frac{4B_k}{D}.$$

For $J > 0$, putting $p = C/J$ gives

$$0 \le p - \widehat{q}_k \le \min\left\{p, \frac{4B_k + 2}{J+2}\right\}.$$

Since $x \mapsto \min\{x, c\}$ is 1-Lipschitz on nonnegative reals, with $A = 24k + 2$ the triangle inequality implies

$$|\widehat{q}_k - q| \le \mathbf{2}\{J > 0\}|C/J - q| + \min\left\{q, \frac{A}{J+2}\right\}. \tag{A.22}$$

When $J = 0$, $C = 0$ and $\widehat{q}_k = 0$; since $A \ge 2$, the last term equals q, so the inequality still holds without defining C/J. This is valid for every permitted edit realization. Using Lemma A.2, (A.21), $\beta = 1/M$, and $A \le 26k$ gives

$$R \le \frac{31744}{\sqrt{M}} + 159744\frac{k}{M} \le 191488\max\{M^{-1/2}, k/M\}, \qquad k \ge 1.$$

For $k = 0$ the word is unedited. On $J > 0$, the add-one probability is within $1/(J+2)$ of C/J; on $J = 0$ pay unit loss. Lemma A.1 gives $\Pr(J = 0) \le 16/(M+1)$. For completeness, integer-threshold layer cake gives

$$\mathbb{E}\frac{1}{J+2} \le \int_0^1 \min\left\{1, \frac{16}{(M+1)t}\right\} dt \le \frac{8}{\sqrt{M+1}}.$$

Indeed a nonempty event $J < t^{-1} - 2$ uses threshold $\lceil t^{-1} - 2 \rceil \le t^{-1}$; for the last inequality use $\min\{1, z\} \le \sqrt{z}$ and integrate $t^{-1/2}$. Together with Lemma A.2 this is at most $536/\sqrt{M}$, below the asserted constant. No positional message or raw-window input is used in either budget branch. □

B Observation separation and lower bounds

Throughout this section a word has symbols $X_0, \dots, X_{M+1}$, pair states $Y_i = (X_i, X_{i+1})$, and departures $0 \le i < M$. Write $J_{ab} = \sum_{i<M} \mathbf{1}\{Y_i = ab\}$, $J_{\text{cross}} = J_{01} + J_{10}$, and $N_{abc} = H_{abc}(X)$. Thus Y_M is a terminal pair, not a departure. All risks below include independent predictor randomness. The observation $\mathcal{T}_r$ consists of the eight triple counts and the actual first and last r symbols.

B.1 An exact occupation mixture

For $0 < a, t < 1$, let Q_a be the stationary law with pair transition matrix, in the order $00, 01, 10, 11$,

$$P_a = \begin{pmatrix} 1-a & a & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & t & 1-t \end{pmatrix}, \qquad \Pi_a = \frac{(t/a, 2t, 2t, 1)}{1 + t/a + 4t}.$$

In particular the stationary initial factor is $\Pi_a(11) = a/[t + (1+4t)a]$.

Lemma B.1 (Occupation-mixture domination)**.** *Put $F = \{Y_0 = 11, Y_M = 00\}$ and $U = N_{001}$. For every integer j, every integer $u_0 \ge 1$, every interval $I \subset (0,1)$, and every additional event G (possibly depending on a),*

$$\int_I Q_a(F, J_{00} = j, U \ge u_0, G)\, \frac{da}{a} \le \frac{t}{u_0}.$$

Proof. Delete every transition departing from 00, both loops and exits. For $J_{00} = j$, exactly $L = M - j$ transitions remain. Their last transition is $10 \to 00$. Before that transition the retained path is on $B = 01, C = 10, D = 11$. A retained $C \to 00$ followed by an erased exit $00 \to B$ is represented by a marked $C \to B$ of weight $1/2$; a direct $C \to B$ is unmarked and also has weight $1/2$. The erased exit's factor a is restored separately. All other retained transitions keep their probabilities. The marked transition matrix is therefore

$$K(z) = \begin{pmatrix} 0 & 1/2 & 1/2 \\ (1+z)/2 & 0 & 0 \\ 0 & t & 1-t \end{pmatrix}.$$

For $L \ge 1$, define

$$h_{L,u} = \tfrac{1}{2}[z^u](K(z)^{L-1})_{D,C},$$

and set $h_{L,u} = 0$ for $L \le 0$ or $u < 0$. The final transition $C \to 00$ has weight $1/2$ and is not marked. A skeleton with u marks has u completed visits to 00 and one terminal visit. Distributing its $j - u$

loops among these $u+1$ holding slots gives exactly $\binom{j}{u}$ weak compositions. Each composition and marked skeleton recover one word, and deletion recovers both uniquely. Consequently, for $0 \le u \le j$,

$$Q_a(F, J_{00} = j, U = u) = \Pi_a(11)\binom{j}{u} a^u (1-a)^{j-u} h_{M-j,u}. \tag{B.1}$$

Infeasible cases have zero mass. The terminal holding slot is present even though Y_M is not a departure; there is no initial 00 slot.

For $u \ge 1$, the beta integral gives $\binom{j}{u}\mathrm{B}(u, j-u+1) = 1/u$. Thus, if $A_{j,u}$ has the $\mathrm{Beta}(u, j-u+1)$ law,

$$\int_0^1 Q_a(F, J_{00} = j, U = u)\,\frac{da}{a} = \frac{h_{M-j,u}}{u}\mathbb{E}\Pi_{A_{j,u}}(11) \le \frac{h_{M-j,u}}{u}. \tag{B.2}$$

The expectation is part of the equality: stationarity has not been cancelled. For $u = 0$ the corresponding factor is instead $h_{M-j,0}\int_0^1 (1-a)^j/[t + (1+4t)a]\,da$, and no $1/u$ formula is used. This is why $u_0 > 0$ is required.

To sum over marks at the exact outside horizon, put $R = K(1)$. This matrix is stochastic. Started at D, let b_n, c_n, d_n be its probabilities of B, C, D. Then

$$c_0 = 0, \quad c_1 = t, \quad b_n = c_{n-1}\ (n \ge 1), \qquad c_n = \tfrac{1}{2}c_{n-2} + t d_{n-1}\ (n \ge 2).$$

Induction gives $c_n \le 2t$. Hence, for each $L \ge 1$,

$$\sum_{u \ge 0} h_{L,u} = \tfrac{1}{2}c_{L-1} \le t.$$

First drop G and enlarge I to $(0,1)$ in the nonnegative integral. Then sum (B.2) over $u \ge u_0$ to obtain

$$\int_I Q_a(F, J_{00} = j, U \ge u_0, G)\,\frac{da}{a} \le \sum_{u \ge u_0} \frac{h_{M-j,u}}{u} \le \frac{1}{u_0}\sum_{u \ge 0} h_{M-j,u} \le \frac{t}{u_0}.$$

No conditioning on a prescribed number of outside marks or replacement of the horizon $M - j$ is involved. □

B.2 Relocating loops while preserving sampled windows

Lemma B.2 (Interior loop relocation)**.** *Suppose $Y_0 = 11, Y_M = 00$, $2 \le r$, and $2r \le M + 2$. Put $l = r - 2$, $h = M - r + 2$. If the interior vertices $Y_l, \dots, Y_h$ contain a visit to 11 strictly before a visit to 00, there is a deterministic transformation C_r preserving the complete triple histogram and the actual first and last r symbols, such that, with $j = J_{00}$,*

$$d_H(C_r(X), 1^{M+2-j}0^j) \le 3J_{\mathrm{cross}} + 4r - 5. \tag{B.3}$$

If at least one of the two movable diagonal-loop types is absent, the same conclusion holds without the ordered-visit condition.

Proof. Freeze departures $i < l$ and $i \ge h$. These are $2r - 4$ edges and spell exactly the sampled boundary symbols. From the interior path remove only the 111 and 000 loops, in numbers A_1 and A_0, retaining every other edge in order. Let $S_0, \dots, S_L$ be its vertices. Deleting self-loops neither removes a distinct-state visit nor reverses its order. Choose the first retained 11 and the first subsequent retained 00; insert all A_1 loops at the former and all A_0 at the latter. The choice is

deterministic. Deleting or inserting a loop preserves its attachment pair. Reinserting exactly the removed number of edges preserves the absolute suffix attachment index h, so the frozen edge labels and all sampled boundary symbols are unchanged. Every triple label is retained or reinserted once, proving histogram preservation. Unfrozen boundary-crossing triples need not keep their individual positions.

Here is the required positional count argument. A binary word of length N with a distinguished block of A_1 ones before a distinguished block of A_0 zeros has o other ones and z other zeros. If $j = A_0 + q$, $q \geq 0$, then

$$d_H(\text{word}, 1^{N-j}0^j) \leq \max\{z + q,\, 2o + z - q\}. \tag{B.4}$$

Indeed, write z_- for zeros before the target cut. The exact mismatch count is $2z_- + j - (A_0 + z)$. If a distinguished one lies after the cut, all distinguished zeros lie after the cut and $z_- \leq z$, giving $z + q$. Otherwise every one after the cut is among the o other ones; writing o_+ for their number, the other exact identity is $2o_+ + (A_0 + z) - j$, at most $2o + z - q$. This proof does not presume any order among the remaining symbols.

Put $u = N_{001}$, $v = N_{011}$, $p = N_{010}$, $w = N_{101}$, and $d = J_{\text{cross}}$. Endpoint flow gives

$$N_{100} = u + 1, \qquad N_{110} = v + 1, \qquad d = p + v + u + 1 + w.$$

Let f_1, f_0 be the frozen $111,000$ loop counts, so $f_1 + f_0 \leq 2r - 4$. The initial pair contributes two other ones. Counting the appended symbol of every remaining edge gives exactly

$$\begin{aligned} o &= 2 + u + v + w + f_1 = d + 1 - p + f_1, \\ z &= p + u + v + 2 + f_0 = d + 1 - w + f_0, \\ q &= j - A_0 = u + f_0 \geq 0. \end{aligned}$$

Since $u \leq d - 1$ and $d \geq 1$,

$$\begin{aligned} z + q &= d + 1 - w + u + 2f_0 \leq 2d + 2f_0, \\ 2o + z - q &= 3d + 3 - 2p - w - u + 2f_1 \leq 3d + 3 + 2f_1. \end{aligned}$$

Each is at most $3d + 4r - 5$, proving (B.3). If only one movable loop type occurs, put its block at any retained visit to its state, which exists because the loops were deleted there. The same two-block argument treats the absent block as empty; if both are absent use the identity transformation.

Finally, j is the sampled word's actual departure count. The comparison word has instead

$$J_{00}(1^{M+2-j}0^j) = j - 2 \quad (2 \leq j \leq M). \tag{B.5}$$

For this range its final pair and the transformed word's final pair are both 00, so all their Hamming differences are in the editable first M coordinates. □

B.3 A finite matched-observation lower bound

Proposition B.3 (Finite critical lower bound)**.** *Let*

$$c = 2^{-16}, \quad A = 2^{12}, \quad t = \frac{ck}{M}, \quad u_0 = \lfloor ck/4 \rfloor, \quad T = \lfloor M/8 \rfloor, \quad L_- = \log \frac{1}{4At}.$$

Assume integer M, k, r satisfy

$$\begin{gathered} M \geq 64, \quad 2 \leq r \leq k/128, \quad k \leq M/8, \quad ck \geq 1024, \\ At < 1/4, \qquad \frac{M}{k^2} \leq \frac{Ac^2}{2^{20}}. \end{gathered} \tag{B.6}$$

For any fixed predictor based on $\mathcal{T}_r$, let S be its supremum clean risk over the stationary laws Q_a, $At \le a \le 1/4$. Every such source satisfies the all-state certificate with supplied $\beta = 1/M$. There is a stationary source with the same supplied certificate and one source-independent strict-k corruption kernel for which

$$R_{\mathrm{null}} \ge \frac{u_0}{8Mt}\left[\left(\frac{t}{16} - S\right) L_-\right]_+ . \tag{B.7}$$

The kernel protects the final pair on every input and every output.

Proof. We give the source, probability, and kernel calculations separately.

Source legality and all sixteen certificate entries. Throughout $t \le c/8$, $a \ge At \ge t$, and $a \le 1/4$. Set $D = 1 + t/a + 4t$. Then $1 \le D \le 1 + 1/A + c/2 < 4/3 < 2$. The flow equalities $\Pi_a(00)a = \Pi_a(10)/2$ and $\Pi_a(11)t = \Pi_a(01)/2$, together with $\Pi_a(01) = \Pi_a(10)$, verify stationarity directly. Multiplication gives

$$P_a^2 = \begin{pmatrix} (1-a)^2 & (1-a)a & a/2 & a/2 \\ 1/4 & 1/4 & t/2 & (1-t)/2 \\ (1-a)/2 & a/2 & 1/4 & 1/4 \\ t/2 & t/2 & (1-t)t & (1-t)^2 \end{pmatrix}. \tag{B.8}$$

Every displayed entry is at least $t/2$: use $a \ge t$ and $a, t \le 1/4$ for the products and all other entries are immediate. Since $t/2 = ck/(2M) \ge 1/M$, every starting and destination state obeys $P_a^2(u, v) \ge M^{-1}\Pi_a(v)$. Also $a \ge At \ge 2/M$, so $a - \beta \ge a/2$.

Common-atom covariance and coupling bounds. The four paths to 11 of length three, from the states in order, have probabilities

$$(1-a)a/2, \qquad (1-t)^2/2, \qquad a/4, \qquad (1-t)^3.$$

These arise respectively from paths $00, 00, 01, 11$; $01, 11, 11, 11$; $10, 00, 01, 11$; and $11, 11, 11, 11$. Each is at least $\lambda = a/4$, so $P_a^3(x, \cdot) \ge \lambda\delta_{11}$ for every x. Split each three-step transition into this common atom and a residual probability law. Shared independent Bernoulli trials of probability λ couple any two chains at the first common atom. Fill in each three-step block by its conditional bridge; this preserves the full one-step path law. After meeting, use identical transitions. Thus the contraction coefficient at lag h is bounded by

$$\alpha_h = (1 - \lambda)^{\lfloor h/3 \rfloor}.$$

For a stationary chain W, a departure-state-set count with stationary mass p, or a count of a specified directed edge with stationary probability p, satisfies

$$\mathrm{Var}(V) \le \frac{7Mp}{\lambda}. \tag{B.9}$$

For an edge indicator I_i and $h \ge 1$, condition on the state after edge i. The conditional probability of edge $i + h$ is a $[0, 1]$-valued function after $h - 1$ transitions, whose deviation from its stationary value is at most α_{h-1}. Therefore $|\mathrm{Cov}(I_i, I_{i+h})| \le p\alpha_{h-1}$. For a state-set indicator use $\alpha_h \le \alpha_{h-1}$ instead. Since $\sum_{h \ge 1} \alpha_{h-1} = 3/\lambda$, expansion of the variance gives $Mp(1 + 6/\lambda) \le 7Mp/\lambda$, as claimed.

Couple a chain Z started at 00 to stationary W by the same block construction. Its meeting boundary σ has

$$\mathbb{E}\sigma \le 3/\lambda = 12/a.$$

Every departure-state or edge count in the two paths differs by at most σ pathwise: after boundary σ the paths agree, and there are at most σ earlier departures. No independence of σ and the stationary counts is needed.

Stationary reversal and count corrections. Let $\mathcal{R}(ab) = ba$. The stationary triple identity

$$\Pi_a(ab)P_a(ab, bc) = \Pi_a(cb)P_a(cb, ba)$$

holds for every triple: the only nonpalindromic cases are $001, 100$ and $011, 110$, given by the two flow identities above. It implies $\mathcal{R}P_a^*\mathcal{R} = P_a$, where P_a^* is the stationary reverse pair kernel. Reversing the symbols of a stationary word thus preserves its law. In particular, with $F = \{Y_0 = 11, Y_M = 00\}$ and $E = \{Z_M = 11\}$,

$$Q_a(F) = \Pi_a(00)P_{00}(E),$$

and conditional on F, the reversed pair path $Z_i = \mathcal{R}Y_{M-i}$ has law $P_{00}(\cdot \mid E)$. For $M \geq 64$, $\lfloor M/3 \rfloor \geq M/4$, and therefore

$$\alpha_M \leq e^{-aM/16} \leq \frac{16}{aM} \leq \frac{16}{Ack} \leq \frac{1}{4}.$$

Since $\Pi_a(11) = 1/D \geq 3/4$, we have $P_{00}(E) \geq 1/2$.

Put $V(W) = N_{001}(W)$ and $x = ck$. Its stationary mean is $Mt/D \in [x/2, x]$. Consider the four conditions

$$\sigma \leq x/16, \quad |V(W) - Mt/D| \leq x/8, \quad J_{\text{cross}}(W) \leq k/32, \quad J_{00}(W) \leq M/16. \tag{B.10}$$

Their failure probabilities, respectively, are at most

$$\frac{192M}{Ac^2k^2}, \quad \frac{1792M}{Ac^2k^2}, \quad \frac{458752M}{Ak^2}, \quad \frac{16}{A}. \tag{B.11}$$

The first follows from $\mathbb{E}\sigma \leq 12/a$ and $a \geq Ack/M$. For the second, (B.9) with $p = t/D$ gives $\operatorname{Var} V \leq 28Mt/a$; divide by $(ck/8)^2$. For the third, the mean is $4Mt/D \leq 4ck \leq k/64$, and $\operatorname{Var} J_{\text{cross}} \leq 112Mt/a \leq 112M/A$; divide by $(k/64)^2$. For the last use $\Pi_a(00) = t/(aD) \leq 1/A$ and Markov's inequality. Conditioning the joint coupling on E costs at most two. Under (B.6), twice the sum in (B.11) is bounded by

$$2\left(\frac{1984}{2^{20}} + \frac{458752c^2}{2^{20}} + \frac{16}{A}\right) < \frac{1}{16}.$$

On F, the exact reversal identities are

$$\begin{aligned} J_{00}(Z) &= J_{00}(Y) + 1, \\ J_{\text{cross}}(Z) &= J_{\text{cross}}(Y), \\ N_{001}(Z) &= N_{100}(Y) = N_{001}(Y) + 1. \end{aligned} \tag{B.12}$$

The first compares arrival indices $1, \ldots, M$ with departure indices $0, \ldots, M-1$; the second loses no mixed endpoint; the third combines triple reversal with entry-minus-exit flow at 00. Thus (B.10), the pathwise coupling bound, and $x \geq 1024$ imply for the original path

$$\begin{aligned} U &\geq x/2 - x/8 - x/16 - 1 \geq x/4 \geq u_0, \\ J_{\text{cross}} &\leq k/32 + ck/16 \leq k/16, \\ 2 \leq u_0 \leq J_{00} &\leq M/16 + ck/16 \leq M/8. \end{aligned}$$

The integer upper count bound is $J_{00} \leq T$.

Ordered visits in the interior. Let $l = r - 2$ and $h = M - r + 2$. Since $r \le M/1024$, $h - l = M - 2r + 4 \ge M/2$. Write B for a visit to 11 before a later visit to 00 within $Y_l, \dots, Y_h$. Because $M - h = l$ and $M - l = h$, reversal sends this event precisely to a visit to 00 before a later visit to 11 within $Z_l, \dots, Z_h$.

The ordinary expected hitting times of 00 and 11, zero at their targets and listed in state order, are

$$h_{00} = (0,\, 4 + 1/t,\, 3 + 1/(2t),\, 3 + 3/(2t)),$$
$$h_{11} = (3 + 3/(2a),\, 3 + 1/(2a),\, 4 + 1/a,\, 0).$$

For completeness, the first vector follows by solving $h_{01} = 1 + (h_{10} + h_{11})/2$, $h_{10} = 1 + h_{01}/2$, and $h_{11} = 1 + t h_{10} + (1 - t)h_{11}$, with $h_{00} = 0$. For the second solve $h_{00} = 1 + (1 - a)h_{00} + a h_{01}$, $h_{01} = 1 + h_{10}/2$, and $h_{10} = 1 + (h_{00} + h_{01})/2$, with $h_{11} = 0$. On $a, t \le 1/4$ the two maxima are $3 + 3/(2t)$ and $3 + 3/(2a)$. From any state at time l, hitting 00 and then 11 therefore has expected additional time at most

$$6 + \frac{3}{2t} + \frac{3}{2a} \le \frac{6}{t}.$$

This uses the strong Markov property at the first hit: partition by its finite time and state, apply the usual Markov property on each event, and sum. Distinct targets make the ordering strict. Markov's inequality over $h - l \ge M/2$ bounds the unconditioned failure probability by $12/(tM)$. Dividing by $P_{00}(E) \ge 1/2$ gives

$$Q_a(B^c \mid F) \le \frac{24}{ck} \le \frac{3}{128}.$$

This estimate is uniform in the time-l state; it assumes no independence from the final state or the counts.

Let G be the intersection of B with $U \ge u_0$, $J_{\mathrm{cross}} \le k/16$, and $2 \le J_{00} \le T$. The preceding count and hitting bounds give $Q_a(G \mid F) \ge 1 - 1/16 - 3/128 > 1/2$. Retaining the rare endpoint mass,

$$Q_a(F \cap G) \ge \Pi_a(00)/4, \qquad (a - \beta) Q_a(F \cap G) \ge t/16. \tag{B.13}$$

Indeed $P_{00}(E) \ge 1/2$, $a - \beta \ge a/2$, and $a\Pi_a(00) = t/D \ge t/2$.

The null and its full word masses. Take a stationary first-order binary source with $T_*(1,0) = q_* = 1/(2M)$ and $T_*(0,1) = \beta = 1/M$. Its symbol masses are $\pi_*(0) = 1/3$, $\pi_*(1) = 2/3$. Its stationary pair law is $\Pi_*(s,z) = \pi_*(s) T_*(s,z)$, and for all starting and destination pair states,

$$\frac{P_*^2((u,v),(s,z))}{\Pi_*(s,z)} = \frac{T_*(v,s)}{\pi_*(s)}.$$

For $(v,s) = (0,0), (0,1), (1,0), (1,1)$ these four ratios are $3(1-\beta)$, $3\beta/2$, $3q_*$, $3(1-q_*)/2$, respectively. They are all at least $\beta + q_* \ge \beta$, checking every pair-state certificate entry. The target at terminal 00 is Bernoulli(β), not Bernoulli(q_*). For $2 \le j \le T$ put $x^{(j)} = 1^{M+2-j}0^j$. These disjoint words have full stationary masses

$$v_j = \frac{2}{3}(1 - q_*)^{M+1-j} q_* (1 - \beta)^{j-1} \ge \frac{1}{8M}. \tag{B.14}$$

Bernoulli's inequality gives the prefix factor at least $1/2$ and the suffix factor at least $7/8$, so in fact the product is at least $7/(48M) > 1/(8M)$. These masses include the initial symbol and every one of the $M + 1$ transitions.

A predeclared strict kernel with identity rejection. For $2 \le j \le T$, define a finite measure on words by

$$\nu_j(dy) = \int_{At}^{1/4} Q_a(dy; F, U \ge u_0, J_{00} = j)\, \frac{da}{a}, \qquad \mu_j = \nu_j(\text{all words}).$$

Lemma B.1 gives $\mu_j \le t/u_0$. For $\mu_j > 0$, normalization ν_j/μ_j is a probability law. These laws depend only on declared M, k, r and the auxiliary family, not the true source, its parameters, its initial law, or a predictor seed.

On input $x^{(j)}$ with $\mu_j > 0$, draw Y once from ν_j/μ_j. If B holds, form the deterministic $C_r(Y)$ of Lemma B.2. Accept it only if its actual Hamming distance from $x^{(j)}$ is at most k and its final pair equals that of $x^{(j)}$. Otherwise output the unchanged input. On zero-mass labels and all other input words, also output the unchanged input. This defines one source-independent kernel on all words, with a strict pointwise budget and protected final pair, not merely an average budget. It uses one draw, not repeated sampling until acceptance.

On G, Lemma B.2 bounds the distance by $3k/16 + 4r \le 3k/16 + k/32 < k$, so every check succeeds, and

$$\mathcal{T}_r(C_r(Y)) = \mathcal{T}_r(Y).$$

The same statistic identity holds on every transformed acceptance; it is not asserted on identity rejections. These are the sampled alternative's boundary windows, not necessarily the null input's windows. Input-window edits are allowed and included in the Hamming check. The suffix label obeys exactly $J_{00}(x^{(j)}) = j - 2$ and $J_{00}(Y) = j$.

Nonnegative-loss ordering. For a sampled word y, let $\ell_a(y)$ be the predictor's expected absolute error against a on $\mathcal{T}_r(y)$, and let $\ell_*(y)$ be its corresponding error against β. On F the clean actual terminal target is a; the triangle inequality, also after averaging independent decoder randomness, gives $\ell_*(y) \ge a - \beta - \ell_a(y)$. For $\mu_j > 0$, (B.14) gives

$$\frac{v_j}{\mu_j} \ge \rho := \frac{u_0}{8Mt}.$$

First discard nonnegative losses from other null inputs and rejected proposals. On the remaining nonnegative accepted losses, replace v_j/μ_j by ρ. Next restrict to G, where acceptance and statistic matching hold. Only then use the signed triangle bound:

$$\begin{aligned} R_{\text{null}} &\ge \rho \int_{At}^{1/4} \mathbb{E}_{Q_a}[\mathbf{1}_{F\cap G}\ell_*(Y)]\, \frac{da}{a} \\ &\ge \rho \int_{At}^{1/4} \left((a-\beta)Q_a(F\cap G) - \mathbb{E}_{Q_a}[\mathbf{1}_{F\cap G}\ell_a(Y)]\right) \frac{da}{a} \\ &\ge \rho \left(\frac{t}{16} - S\right) L_-. \end{aligned}$$

The restricted alternative loss is bounded by its unconditional clean risk, hence by S. Risk is nonnegative, giving (B.7). In particular, no lower mass multiplier has been placed in front of a possibly negative quantity, and no endpoint-conditioned loss has been substituted for the unconditional target risk. □

B.4 Consequences for widths and supplemental messages

Proposition B.4 (Width and fixed-fraction consequences)**.** *For a prescribed sequence $2 \le r_M \le M+2$, a finite uniform factor relative to $g_M(k) = \max\{M^{-1/2}, k/M\}$ exists if and only if $\liminf_{M\to\infty} r_M/M > 0$, for stationary or arbitrary starts. For each fixed $0 < \alpha \le 1/4096$ and $r_M = \max\{2, \lceil \alpha M \rceil\}$,*

$$2^{-26} \log \frac{1}{32\alpha} \le C_{\mathrm{stat}}(\alpha) \le C_{\mathrm{all}}(\alpha) \le 2^{19} \left(1 + \log \frac{2}{\alpha}\right).$$

With exact counts and protected terminal pair, optionally also the initial pair, a post-corruption full-word encoder needs at least two supplemental symbols to attain a finite uniform factor; two suffice when its message is exact and fault-free.

Proof. Suppose first that the liminf is zero. Select an increasing subsequence on which $r_M/M \to 0$, and on that subsequence put

$$b_M = \max\{r_M, \sqrt{M}\}, \qquad k_M = \lfloor \sqrt{M b_M} \rfloor.$$

Then $b_M = o(M)$, $k_M/M \to 0$, $k_M/b_M \to \infty$, $r_M/k_M \to 0$, and $k_M/\sqrt{M} \to \infty$. Moreover $M/k_M^2 \to 0$, so every condition in (B.6) holds eventually. Here $t_M = ck_M/M$, $\rho = u_0/(8Mt_M)$, $L_- \to \infty$, and $\rho \to 1/32$. If one k-blind family had a finite admissible factor C, its clean risk on every alternative at this same M and supplied certificate would satisfy $S_M \le C/\sqrt{M} = o(t_M)$. For example, eventually $\rho \ge 1/64$ and $t_M/16 - S_M \ge t_M/32$, so

$$\frac{R_{\mathrm{null}}}{g_M(k_M)} \ge \frac{c}{2048} L_- \longrightarrow \infty,$$

since $g_M(k_M) = k_M/M$ eventually. This contradicts C. The alternatives may depend on the tested budget, but the predictor and its clean guarantee are the same map at the same horizon. These stationary witnesses also rule out an all-start guarantee.

Conversely, if the liminf is positive, choose $\delta > 0$ and M_0 so $r_M \ge \delta M$ for $M \ge M_0$. Put $m_M = \min\{r_M - 2, \lfloor M/2 \rfloor\}$. Then $(m_M + 1)/M$ is bounded away from zero for all sufficiently large M. The suffix estimate of Proposition A.9 is at most $2^{19}[1 + \log(M/(m_M+1))]g_M(k)$ at $\beta = 1/M$, so gives one uniform all-start factor eventually. A finite initial set of horizons is harmless: any prediction has risk at most one and $g_M(k) \ge M^{-1/2}$. This proves the exact criterion without monotonicity of the widths.

Now fix $\alpha > 0$ in the stated interval before sending M to infinity. For any family with finite admissible stationary factor C, put $k = 128 r_M$ at sufficiently large horizons. Then

$$\frac{k}{M} \to 128\alpha \le \frac{1}{32}, \quad t \to 128c\alpha > 0, \quad At \to 8\alpha < \frac{1}{4},$$

all finite-domain conditions eventually hold, and

$$L_- \to \log \frac{1}{32\alpha}, \qquad \frac{u_0}{8Mt} \to \frac{1}{32}, \qquad S/t \le \frac{C}{t\sqrt{M}} \to 0.$$

For these budgets $k \ge \sqrt{M}$ eventually. Using the deliberately loose $g_M(k) \le 2k/M$ in (B.7), the positive limiting bracket gives

$$C \ge \liminf_{M\to\infty} \frac{R_{\mathrm{null}}}{g_M(k)} \ge \frac{1}{32}\frac{c}{32} \log \frac{1}{32\alpha} = 2^{-26} \log \frac{1}{32\alpha}.$$

This holds for every admissible finite factor, so taking the infimum does not require an attained optimum. The upper follows from Proposition A.9: $m_M + 1 \geq \alpha M/2$, including $r_M = 2$, hence $1 + \log(M/(m_M + 1)) \leq 1 + \log(2/\alpha)$. The stationary class is a subclass of the arbitrary-start class. These arguments take the horizon limit for each fixed fraction and factor; they neither exchange limits nor give an every-budget logarithmic risk lower bound.

Finally, a one-symbol supplemental alphabet is constant, so leaves only counts and the supplied endpoint pair or pairs. At $r_M = 2$ even both endpoint pairs are insufficient by the necessity just proved. The exact one-bit factorization in Proposition A.9 supplies two symbols and has all-start risk at most $(2^{17} + 2^{23})g_M(k)$. This concerns a fault-free message computed by scanning the whole already corrupted word, not one raw coordinate or a message subject to further corruption. □

B.5 Elementary same-information oracle lower floors

Lemma B.5 (Critical oracle lower floors)**.** *For every $M \geq 64$ and $0 \leq k \leq M$, the stationary minimax risk with supplied $\beta = 1/M$ is at least $g_M(k)/96$, even if the predictor knows k and observes the entire corrupted word. Consequently this lower bound holds for each restricted observation considered here and for the arbitrary-start minimax risk.*

Proof. We give separate clean and edit witnesses; taking their maximum is permitted in a minimax lower bound.

Clean floor. Put $N = M + 2$ and use stationary IID Bernoulli sources

$$p_\pm = \tfrac{1}{2} \pm \frac{1}{16\sqrt{N}}.$$

Their pair chains satisfy $P^2(u, v) = \Pi(v)$ for every row, hence the supplied certificate. The target is $p_\pm$ at every terminal pair. The elementary Bernoulli divergence bound

$$d(p\|q) \leq \frac{(p-q)^2}{q(1-q)}$$

follows by applying $\log x \leq x - 1$ to its two summands. With $\Delta = p_+ - p_- = 1/(8\sqrt{N})$, it gives one-letter divergence at most $1/(8N)$. Independence makes the full-word divergence at most $1/8$, and their total variation distance is at most $1/4$. For clarity, this last inequality follows without a statistical independence assertion about a selected row: for finite probability laws P, Q, group their masses on $A = \{x : P(x) \geq Q(x)\}$ and its complement. The log-sum inequality (Jensen's inequality for $x \log x$) gives $D(P\|Q) \geq d(P(A)\|Q(A))$. For fixed q, the second derivative in p of $d(p\|q)$ is $1/[p(1-p)] \geq 4$; its value and first derivative vanish at $p = q$. Thus $D(P\|Q) \geq 2(P(A) - Q(A))^2 = 2\,\mathrm{TV}(P, Q)^2$.

Testing by the nearer target makes each error incur prediction loss at least $\Delta/2$. Every possibly randomized test has average error at least $(1 - \mathrm{TV}(P, Q))/2$, since the difference of its acceptance probabilities is at most total variation. Thus the average prediction risk is at least

$$\frac{\Delta}{4}(1 - \mathrm{TV}(P, Q)) \geq \frac{3}{128\sqrt{N}} \geq \frac{1}{64\sqrt{M}}.$$

Identity corruption is allowed at every k, including for a known-budget predictor.

Edit floor. For $k \geq 1$, use the uniform prior on IID Bernoulli parameters $p \in [1/4, 3/4]$. Put $b = k + 1$. If the first M symbols contain n ones, retain a uniformly chosen set of $r = b\lfloor n/b \rfloor$ of those ones, turning the others to zero. Leave the final two symbols U unchanged. This single kernel is source-independent and changes $n - r \leq b - 1 = k$ editable bits on every outcome. For a specified retained set T of size r, conditional on n,

$$\Pr(T \text{ retained} \mid n) = \frac{\binom{M-r}{n-r}}{\binom{M}{n}\binom{n}{r}} = \frac{1}{\binom{M}{r}}.$$

Hence conditional on r, U the output arrangement is uniform and independent of p, even after mixing the possible n in a bucket. The full observation factors through at most

$$D = 4(\lfloor M/b \rfloor + 1)$$

labels (r, U) followed by a p-independent channel. The four protected pair possibilities are counted, not conditioned away.

Realize that channel using a seed independent of p and of the label, and likewise use an independent decoder seed. Freeze both seeds. There are at most D possible prediction values. For the uniform prior on an interval of length $\ell = 1/2$, the union of radius-v neighborhoods of those values has prior mass at most $2Dv/\ell$, regardless of how a label is randomly selected conditional on p. Outside that union every possible prediction has error greater than v. Integrating this probability bound gives Bayes absolute-error risk at least

$$\int_0^{\ell/(2D)} \left(1 - \frac{2Dv}{\ell}\right) dv = \frac{\ell}{4D}.$$

Averaging the independent seeds preserves it. Consequently some parameter in the prior support has risk at least

$$\frac{1}{32(\lfloor M/(k+1) \rfloor + 1)} \geq \frac{k+1}{32(M+k+1)} \geq \frac{k}{96M}.$$

IID stationarity and $P^2(u, v) = \Pi(v)$ again give the same source class and supplied certificate, with target p at every row. Combining the clean floor and edit floor yields $g_M(k)/96$, also at $k = 0$. A predictor with less information can only have a larger minimax risk, and allowing arbitrary starts contains the stationary experiment. □

C Simultaneous finite-horizon competitive factors

We prove Theorem 3.1 using the finite estimates proved in the preceding two sections. No limit in M or r is taken. Throughout this section $\ell = 1 + \log((M+2)/r) \geq 1$, the certificate is $1/M$, and Γ denotes the finite-horizon quantities of (2.7), not the sequence factors C.

Proof of Theorem 3.1. **Upper bound, including overlapping windows.** Set $m = \min\{r - 2, \lfloor M/2 \rfloor\}$ and $L = 1 + \log(M/(m+1))$. Since $r - 1 \geq r/2$ and $\lfloor M/2 \rfloor + 1 \geq M/2$,

$$\frac{M}{m+1} \leq \max\left\{\frac{2M}{r}, 2\right\} \leq \frac{2(M+2)}{r}.$$

The last step uses $r \leq M + 2$ and covers full or overlapping windows. Hence $L \leq \ell + \log 2 \leq 2\ell$. Proposition A.9 gives one counts-and-suffix map, independent of k, with all-start risk at most

$2^{19} L g_M(k) \le 2^{20} \ell g_M(k)$ for every k. Stationary experiments are a subclass of arbitrary-start experiments, so taking the infima over the same maps also gives $\Gamma_{\mathrm{stat}}(M,r) \le \Gamma_{\mathrm{all}}(M,r)$.

Lower bound and the small-logarithm case. Fix any allowable map A and write $G = \Gamma_{\mathrm{stat}}(A; M, r) \le \sqrt{M}$. Lemma B.5, at $k = 0$, implies $G \ge 1/96$ even if full history were available. If $\ell < 64$, then $1/96 \ge 2^{-30}\ell$, proving the desired bound. For $\ell \ge 64$, there is nothing to prove when $G \ge 2^{-30}\ell$. Suppose instead

$$G < 2^{-30}\ell. \tag{C.1}$$

As $r \ge 2$ and $M \ge 2$, $(M+2)/r \le M$. Thus $\log M \ge \ell - 1 \ge 63$, and $M > 2^{80}$ because $80 \log 2 < 56 < 63$. Choose the integer budget

$$k = \left\lceil \max\{128r, M^{3/4}\} \right\rceil. \tag{C.2}$$

This budget depends only on (M, r); the same map A is used at this budget and at zero.

Every finite lower-domain condition. Use the constants from Proposition B.3, writing $A_0 = 2^{12}$ for its numerical constant to distinguish it from the map:

$$c = 2^{-16}, \qquad t = ck/M, \qquad u_0 = \lfloor ck/4 \rfloor, \qquad L_- = \log \frac{1}{4A_0 t}.$$

The choice of k gives $r \le k/128$, $k \ge M^{3/4}$, and

$$\frac{M}{k^2} \le M^{-1/2} < 2^{-40} = \frac{A_0 c^2}{2^{20}}, \qquad ck \ge 2^{-16} M^{3/4} > 2^{44} \ge 1024.$$

Moreover

$$\frac{r}{M} = (1 + 2/M) e^{1-\ell} \le 2e^{-63} < 2^{-79}.$$

Since $k \le 128r + M^{3/4} + 1$, this implies

$$\frac{k}{M} < 2^{-72} + 2^{-20} + 2^{-80} < 2^{-19} < \frac{1}{8}.$$

In particular k is a legal budget, $k \le M/8$, and $A_0 t = k/(16M) < 1/128 < 1/4$. Together with the original integer $M \ge 64, r \ge 2$, these verify every condition in (B.6). The windows needed in that proof are nonoverlapping since $r \le k/128 \le M/1024$. Also $ck \ge 1024$ implies $u_0 \ge ck/8$, so

$$\rho := \frac{u_0}{8Mt} \ge \frac{1}{64}. \tag{C.3}$$

The same map's clean risk. Let S be the supremum of the clean risks on the stationary alternatives in Proposition B.3. All qualify at the same supplied certificate, so $S \le G/\sqrt{M}$. For $x \ge 0$, differentiation shows that $(1+x)e^{-x/4}$ is maximized at $x = 3$, with value $4e^{-3/4} < 4$. Thus $\ell \le 1 + \log M \le 4M^{1/4}$ and (C.1) gives the finite inequalities

$$S < 2^{-28} M^{-1/4} \le 2^{-21} M^{-1/4} \le t/32. \tag{C.4}$$

The last step uses $k \ge M^{3/4}$ and $c/32 = 2^{-21}$. No asymptotic small-order estimate or distinct clean estimator is used.

A logarithm uniform in both parameters. Since $4A_0 c = 1/4$, $L_- = \log(4M/k)$. Put $x = \max\{128r, M^{3/4}\} \ge 1$; then $k = \lceil x \rceil \le 2x$, and

$$L_- \ge \min\left\{ \log \frac{M}{64r},\ \log(2M^{1/4}) \right\}.$$

The first term equals $\ell - 1 - \log(1 + 2/M) - \log 64$ and is at least $\ell - 2 - \log 64 \geq \ell - 8 \geq \ell/8$, using $\log(1 + 2/M) \leq 1$, $\log 64 < 6$, and $\ell \geq 64$. The second term is at least $(\ell - 1)/4 \geq \ell/8$. Consequently $L_- \geq \ell/8$.

Common-kernel comparison and infimum. Proposition B.3 supplies a qualifying stationary null and one strict source-independent kernel, preserving the actual final pair, with

$$R_{\text{null}} \geq \rho \left[(t/16 - S)\, L_- \right]_+ .$$

Its full proof above includes source legality, mixture normalization, window matching, and pointwise budget checks. Here the bracket is positive by (C.4). Since $k \geq M^{3/4} \geq \sqrt{M}$, one has $g_M(k) = k/M$. Therefore

$$G \geq \frac{R_{\text{null}}}{g_M(k)} \geq \frac{(1/64)(t/32)(\ell/8)}{k/M} = 2^{-30}\ell,$$

contradicting (C.1). The lower bound thus holds for every map and passes to its infimum without requiring attainment. The witnesses are stationary and hence are also available in the all-start class. This completes the finite-horizon proof. □

For clarity about the sequence consequences, if r_M/M has positive liminf, then ℓ_M is bounded eventually, and finitely many earlier horizons have finite factor; the explicit suffix maps give a single uniform family. If that liminf is zero, ℓ_M diverges on a subsequence, so the finite lower forbids a uniform factor for any family. For fixed α, the finite bounds give the logarithmic order, while the original fixed-α proof in Proposition B.4 retains its stronger $2^{-26}\log(1/(32\alpha))$ lower constant. Neither argument claims a logarithmic risk lower at every individual budget.

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